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Real Sport Performances, Uncharted Virtual Worlds

An Institutional-Strategic Perspective on the Rise of
Virtual Sports Through the Case of Virtual Cycling



What actually are virtual sports? An ontological study

The first part of this dissertation includes the development of a universal definition of *virtual sports*. Understanding the precise conditions that make a sporting discipline *virtual* is important to (1) delineate the innovation studied and to (2) develop insights beyond specific sports and technologies.

Virtual sports are analysed along eight universal dimensions of sports: human, voluntary, competition, physical, cognitive, following, governance, institutionalisation. We conclude that virtual sports fulfil each dimension meaningfully. However, an additional “reality” dimension should also be considered.

Compared to esports and other digitised sport-related activities like exergames and simulations, virtual sports can be understood as a *distinct ontological category* between sports and esports based on their unique configuration of physicality and reality. Virtual sports have high *movement decompression* as its technology allows to realistically and accurately mimic the bodily participation of their “parent” or “traditional” sports’ physicality. Additionally, virtual sports games actively mediate and alter the physical performance via algorithms and virtual-world laws, creating a *distinct physical-digital sporting reality*.



We conclude by defining virtual sports as: *Institutionalised, rule-governed contests of human physical and cognitive skills that operate within a distinct physical-digital interreality that are characterised by voluntary human participation, a niche following, and high movement decompression mimicking a traditional parent sport, made possible through virtuality yet unbound to specific configurations of remoteness, (a)synchrony or representational fidelity.*

Drivers and Barriers of virtual sports adoption at different levels.

The empirical part of this dissertation includes five studies. They cover the domains of digitisation, institutional dynamics and innovation strategies across micro-, meso- and macro-level.

Study 1 focuses on the innovation strategies of the micro- and meso-level of innovation strategies. Based on innovation diffusion theory, clubs segment into Innovators & Early Adopters (17%), Early Majority (44%), and Late Majority & Laggards (39%) based on their adoption(-intention) of virtual cycling as-a-club-service. Our model predicts 76% of clubs' segments based on perceived innovation attributes and managerial determinants, indicating that adoption strategies must initially support positive perceptions, skills, and knowledge over infrastructure.

Study 2 identifies the different beliefs clubs have about virtual cycling as-a-club-service. Clubs' beliefs mirror two institutional logics: a Sport-for-All and result-oriented logic, the latter perceived as most valuable. These different beliefs and logics illustrate how adoption strategies of virtual cycling at meso-level are not one-size-fits-all and require logic-specific tailoring.

Study 3 examines institutional dynamics and strategies at meso- and macro-level. To push adoption and gain legitimacy, (inter)national governing bodies employ an institutional strategy (normative framing around safety, reduced costs, inclusivity, and targeting new markets). While a technical strategy (developing features) is lacking, we conclude that both legitimisation strategies are equally important for diffusion.

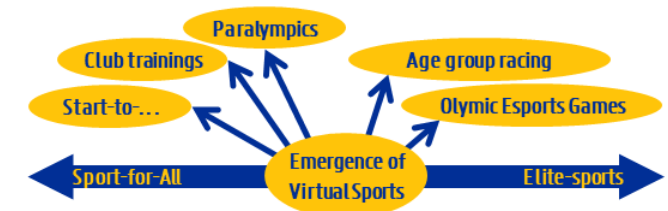
Study 4 illustrates four governance issues that pertain virtual cycling, spanning micro-, meso- and macro-level. (1) (mechanical) doping, (2) platform fragmentation, (3) lacking institutional transparent structures, and (4) lacking support of federations. Changes are needed in virtual cycling's digitisation and institutionalisation, while balancing commercial interests.

Study 5 evidences how algorithms actively determine outcomes in competition. Physical performance (W/kg) significantly better predicts race results on (time) MyWhoosh ($R^2=0.87$) versus Zwift ($R^2=0.76$). To resolve consequent potential governance issues at micro-, meso- and macro-level, dual governance structures and scrutability-without-disclosure models are needed.

Opportunities for and issues challenging virtual sports' legitimacy

When aggregating these results, this dissertation discusses the complex interplay between different domains and different levels of analysis. The (1) digitisation aspects of virtual cycling impact the beliefs, adoption of and use cases of athletes, club volunteers, (inter)national federations, platform providers, and sector-wide standards and regulatory networks. The (2) institutional dynamics in the (e)sports sector impacts all stakeholders. Lacking institutionalised structures negatively affect athletes, organisational constraints make federations dependant on commercial providers, while (inter)national federations must manage two competing policy trajectories: one in sport-for-all and a result-oriented one. The (3) innovation strategies of virtual cycling require significant tailoring towards individuals' perceptions and attitudes, towards the managerial capacities of organisations, and towards the norm-compliant narratives around the ontology of sports: what sport *is*, what it should *be*, and how virtual disciplines support that development.

Each configuration within this complex interplay is fundamental to understand for the twofold policy development and future implementation of virtual cycling: one focussing on recreational sport-for-all and one focussing on results-oriented elite sports. These are explained through institutional theory as different deinstitutionalisation-reinstitutionalisation trajectories, each requiring a distinct approach, having distinct issues, but also offering different opportunities for the sports sector.



This thesis concludes that the process by which virtual sports gains legitimacy and acceptance by society as a new sporting discipline, is one of many technical *and* non-technical challenges. In its future development, a clear vision, strategic planning, dedicated resources, good governance, close collaboration, and interdisciplinary research are indispensable.